Lab 15 Debugging Report: HTTP, JSON & Permissions

Student: _____ ID: _____ App: NewsReader (com.vlu.newsreader), Pixel 6 emulator, API 34-class image. Traces copied from Logcat (tag AndroidRuntime).

Error 1: android.os.NetworkOnMainThreadException

How I reproduced it. I called `HttpClient.getBlocking(url)` directly in `MainActivity.onCreate` (launched with `-e debug mainthread`) instead of going through `lifecycleScope.launch` + `withContext(Dispatchers.IO)`.

```
Process: com.vlu.newsreader, PID: 2524
java.lang.RuntimeException: Unable to start activity ComponentInfo{com.vlu.newsreader/com.vlu.newsreader.MainActivity}
Caused by: android.os.NetworkOnMainThreadException
■at android.os.StrictMode$AndroidBlockGuardPolicy.onNetwork(StrictMode.java:1737)
■at java.net.Inet6AddressImpl.lookupHostByName(Inet6AddressImpl.java:115)
■at com.android.okhttp.internal.huc.HttpURLConnectionImpl.getResponseCode(HttpURLConnectionImpl.java:542)
■at com.android.okhttp.internal.huc.DelegatingHttpsURLConnection.getResponseCode(DelegatingHttpsURLConnection.java:106
■at com.android.okhttp.internal.huc.HttpsURLConnectionImpl.getResponseCode(HttpsURLConnectionImpl.java:30)
■at com.vlu.newsreader.HttpClient.getBlocking(HttpClient.kt:21)
```

Root cause. The main thread draws the screen and handles touches. A network call can take seconds, so Android's StrictMode throws on purpose when a socket or DNS lookup starts on that thread (see `StrictMode$AndroidBlockGuardPolicy.onNetwork` at the top of the *Caused by* chain). The crash happens at the DNS lookup, before any data moves. **Fix:** run the request in `withContext(Dispatchers.IO)`; the app's normal path does this and never blocks the UI.

Error 2: java.lang.SecurityException: Permission denied (missing INTERNET permission?)

How I reproduced it. I built the `noInternet` build type, whose manifest overlay removes `android.permission.INTERNET` (`dumpsys package` shows no INTERNET entry). The request is on the I/O thread, so this is not a thread problem.

```
Process: com.vlu.newsreader, PID: 2607
java.lang.SecurityException: Permission denied (missing INTERNET permission?)
■at java.net.Inet6AddressImpl.lookupHostByName(Inet6AddressImpl.java:150)
■at com.android.okhttp.internal.huc.HttpURLConnectionImpl.getResponseCode(HttpURLConnectionImpl.java:542)
■at com.android.okhttp.internal.huc.DelegatingHttpsURLConnection.getResponseCode(DelegatingHttpsURLConnection.java:106
■at com.android.okhttp.internal.huc.HttpsURLConnectionImpl.getResponseCode(HttpsURLConnectionImpl.java:30)
■at com.vlu.newsreader.HttpClient.getBlocking(HttpClient.kt:21)
■at com.vlu.newsreader.HttpClient$get$2.invokeSuspend(HttpClient.kt:12)
■at kotlinx.coroutines.DispatchedTask.run(DispatchedTask.kt:101)
```

Root cause. Network access is a permission that must be declared in the manifest and is granted at install time. Without it the OS refuses to resolve the host (`Inet6AddressImpl.lookupHostByName`) and throws `SecurityException`. It is not an `IOException`, so my `catch (IOException)` does not catch it; it escapes the coroutine and crashes the app. **Fix:** add `<uses-permission android:name="android.permission.INTERNET" />` as a direct child of `<manifest>`.

Also verified

Airplane Mode: tapping Load shows "Cannot reach the server..." (an `IOException` is caught and logged as `NewsReader: network failure`); the process stays alive. Normal run: 100 posts render in the RecyclerView with both the manual (JSONArray) and Gson parsers.